Optimization of Electric Vehicle Autonomy

Routing Markup Language
(Developed Under KT9, WP5)

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1. EXECUTIVE SUMMARY

This document was created for the need to develop a common structure for use in routing problems. With the widespread use of EVs in logistics and transportation, routing has gained importance. For routing, a Markup Language should be created to bring together different works on routing to a common denominator and create a standard. In this respect, Routing Markup Language was created. Routing is finding the optimal path from a starting point to a destination with specific objectives.

Objective Function: The RML supports various objective functions that can be optimized depending on the routing requirements. These include:

1. **Minimum Distance**: Finding the shortest path between two points.
2. **Minimum Time**: Identifying the fastest route, taking into account speed limits or traffic conditions.
3. **Minimum Energy**: Optimizing routes for energy efficiency, crucial for electric vehicles and reducing environmental impact.

Also, unlike other vehicles, EVs have battery and load constraints. These constraints require taking into account conditions such as vehicle range, charging station, and load capacity in routing. In this respect, optimization algorithms also gain importance.

There are different problems in routing. Some of the most common ones include the A* algorithm, which aims to find the shortest or fastest path between starting point A and ending point B, and the Traveling Salesman Problem (TSP), which tries to solve the problem of one or more vehicles visiting all points in a given set once and returning to the starting point. For TSP, methods such as the Savings algorithm are often preferred. With the increasing use of electric vehicles (EVs), the load capacity problem has emerged in routing. It is aimed to solve the routing problem without exceeding the load capacity limit of the vehicles. This class of problems is called the Capacitated Electric Vehicle Routing Problem (CEVRP).

Similarly, the problem that tries to obtain time windowed solutions with capacity is the Capacity Electric Vehicle Routing Problem with Time Windows (CEVRPTW). It is aimed at serving customers at certain time intervals. These problems are essential for both reducing logistics costs and reducing environmental impacts.

2. HISTORY OF CHANGES

Version	Date	Author	Comment
v1.05	**14.10.2025**	Yasin Ünal	1. Schema updated for hybrid vehicle routing (Vehicle, Plan) 2. Dynamic Routing updates for Task 3. Annotation infos
v1.04	**30.03.2025**	Yasin Ünal	1. Schema Updates (Vehicle, Plan and Algorithm) 2. XSLT and version documentation
v1.03	**03.02.2025**	Yasin Ünal	Document and Schema Updates
v1.02	**31.10.2024**	Yasin Ünal	Schema Updates
v1.01	**02.10.2024**	Yasin Ünal, Yunus Sabri Kırca	New Diagram and Tables
v1.0	**31.07.2024**	Yasin Ünal, Yunus Sabri Kırca	CEVRPwTW

3. ROUTING MARKUP LANGUAGE

Routing Markup Language (RML) is a markup language specifically developed for intelligent transportation systems and route optimization problems. Based on XML (Extensible Markup Language), RML aims to provide a standardized data structure that can be used in transportation, logistics, electric vehicle routing, traffic management, and smart city applications. RML is designed to standardize complex processes such as routing and route planning, facilitate data exchange in an understandable format, and ensure system interoperability. By enabling different systems to communicate in a common language, it enhances efficiency in routing processes. The primary goal of RML is to organize and share data used in route planning systematically. In traditional systems, route data is often stored in various formats and processed by different algorithms. RoutingML establishes a common structure to provide the following advantages in routing processes:

- **Standardization:** Ensures interoperability between different routing applications by using a consistent data format.
- **Flexibility:** Provides a flexible data structure that can address various needs related to routing problems.
- **Data Exchange:** Facilitates data sharing across different systems and platforms.
- **Analysis and Optimization:** Structured data allows routing algorithms to operate more efficiently and accelerates analytical processes.
- **Data Integrity:** Ensures all route-related information is complete and clearly defined.
- **General Usability:** Designed to be applicable in different domains such as logistics, transportation, and electric vehicle routing.
- **Compatibility:** Allows seamless integration with other data formats and systems.
- **Extensibility:** Supports the addition of new components to address different problem types.

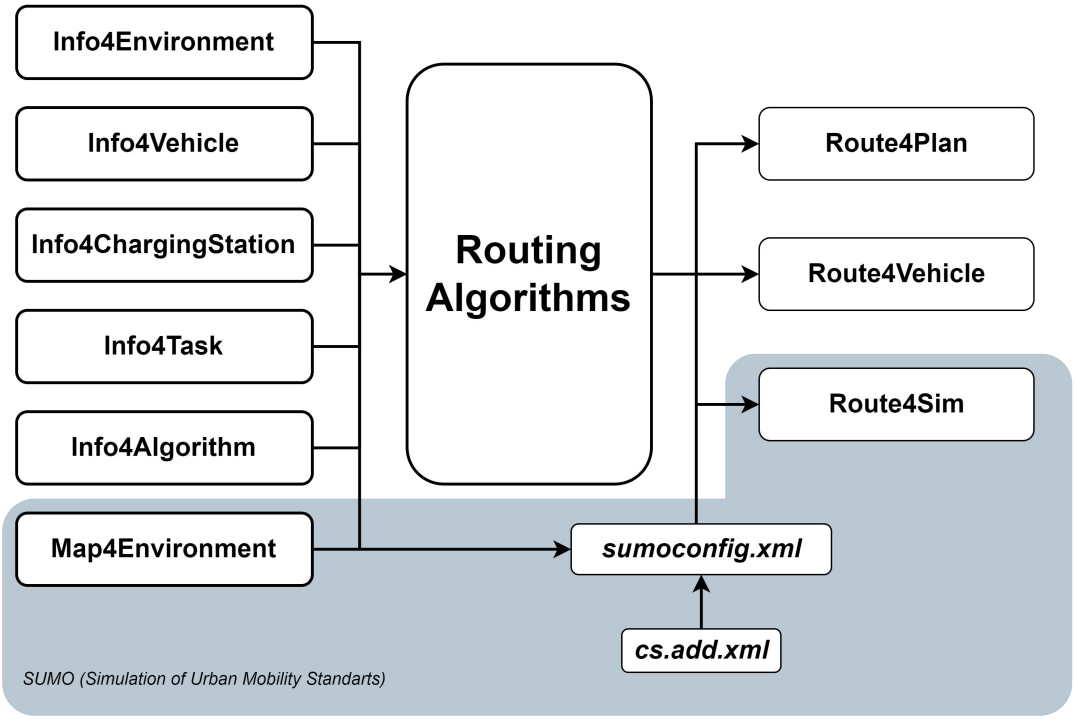

Figure 1 : Routing Markup Language Schema

3.1 Routing Markup Language Overwiew

The Routing Markup Language using an XML-based schema (XSD - XML Schema Definition). This schema is designed to include all fundamental components used in the routing process. The Routing Markup Language uses the black box diagram given in Figure 1 to specify inputs and outputs. The diagram shows a routing system that uses XML-based input files to generate optimized routes. Data collected from Info4CS input files using Info4Environment for environment, Map4Environment for map information, Info4Vehicle for vehicle, Info4Task for task description and OCPP standard for charging stations are given to the routing algorithm. The results are stored in RouteAlgorithm for algorithm analysis, in the map representation RouteSim for export to the SUMO vehicle and in RouteVehicle for the vehicle route.

3.2 Inputs

Tables of the inputs of the algorithm are given in this section.

3.2.1 Info4Environment

Info4Environment contains three classes. These are Environment, LocaleCoordinateCenter and Graph. They are given in Table 1, Table 2 and Table 3 respectively.

Name	Type	Use [minOccurs,

		maxOccurs]
Timezone	xs:string	Required [1,1]
GenerationTime	xs:dateTime	Required [1,1]
UtcOffset	xs:integer	Required [1,1]
Humidity	xs:integer (0-100)	Optional [0,1]
Time	xs:dateTime	Optional [0,1]
Temperature	xs:decimal	Required [1,1]
AirDensity	xs:decimal	Optional [0,1]
Gravity	xs:decimal	Optional [0,1]

Table 1 : Environment fields

Table 2 : LocaleCoordinateCenter

Name	**Element Name**	**Type**	**Use [minOccurs, maxOccurs]**
Id		xs:string	Required (attribute)
Location		xs:complexType	Required [1,1]
	Latitude	xs:decimal	Required [1,1]
	Longitude	xs:decimal	Required

			[1,1]
	Elevation	xs:decimal	Required [1,1]

Table 3 : Graph

Name	Element Name	Element Name	Type	Use [minOccurs, maxOccurs]
Node			xs:complexType	Required [1,unbounded]
	NodeId		xs:string	Required (attribute)
	No		xs:string	Required (attribute)
	Name		xs:string	Required (attribute)
	NodeType		xs:enumeration[xs:string]	Required (attribute)
	Location		xs:complexType	Required
		Latitude	xs:decimal	Required [1,1]
		Longitude	xs:decimal	Required [1,1]
		Elevation	xs:decimal	Optional [0,1]
		X_Coordinates	xs:decimal	Optional [0,1]
		Y_Coordinates	xs:decimal	Optional [0,1]
DistanceMatrix			xs:string	Required [1,1]

EnergyMatrix			xs:string	Required [1,1]
PathMatrix			xs:string	Required [1,1]
LocationMatrix			xs:string	Required [1,1]

You can review Info4Environment.xsd file for definition and details.

3.2.2 Map4Environment

Map4Environment contains OSM_Map. This contains map information retrieved from Open Street Map. This map information is also used for the output format Route4Sim. The structure is given in Table 4.

Table 4 : Map4Environment

Name	Element Name	Element Name	Element Name	Type	Use
OSM_Map				xs:complexType	Required
	Way			xs:complexType	Required
		WayPointId		xs:string	Required
		Name		xs:string	Required
		Location		xs:complexType	Required
			Latitude	xs:decimal	Required
			Longitude	xs:decimal	Required
			Elevation	xs:decimal	Required
		Surface			Optional

You can review Map4Environment.xsd file for definition and details.

3.2.3 Info4Vehicle

Info4Vehicle contains Vehicle, which contains information about the vehicle. It is given in Table 5, 6 and 7.

Table 5. Vehicle Schema

Name	Element Name	Type	Use
ID		xs:string	Required (attribute)
Model		xs:string	Optional (attribute)
Year		xs:integer	Optional (attribute)
MaxSpeed		xs:decimal	Required (attribute)
VehicleMass		xs:decimal	Required (attribute)
MaximumLoadCapacityVolume		xs:decimal	Required (attribute)
MaximumLoadCapacityKg		xs:decimal	Required (attribute)
BatteryCapacity		xs:integer	Required (attribute)
Status		StatusEnum (`Available, InService, Charging, Delivering, Maintenance, Faulted, Unavailable`)	Required [1,1]
CurrentLocation		xs:complexType	Required
	Latitude	xs:decimal	Optional [1,1]
	Longitude	xs:decimal	Optional [1,1]
	Elevation	xs:decimal	Optional [1,1]
CurrentLoadCapacity		xs:complexType	Optional
	Weight	xs:decimal	Optional

			[0,1]
	Volume	xs:decimal	Optional [0,1]
DriverInformation		xs:complexType	Optional
	ID	xs:string	Required [1,1]
	Style [option1, option2, option3]	xs:enumeration [xs:string]	Optional [1,1]

Table 6. EV Type

Name	Element Name	Type	Use
BatteryCapacity		xs:integer	Required (attribute)
SoC		xs:decimal	Required (attribute)
SoH		xs:decimal	Optional (attribute)
Range		xs:decimal	Optional (attribute)
BatteryRechargingRate		xs:decimal	Required (attribute)
EnergyConsumptionRate		xs:decimal	Required (attribute)
ChargingTime		xs:decimal	Optional (attribute)
Front Surface Area		xs:decimal	Optional (attribute)
Torque		xs:decimal	Optional (attribute)
ChargingStrategy		xs:enumeration[xs:	Optional

		string]	[0,1]
	FullCharge	xs:string	Optional [0,1]
	PartialCharge	xs:string	Optional [0,1]
	20-80	xs:string	Optional [0,1]
PropulsionEfficiency		xs:decimal	Optional (attribute)
RecuperationEfficiency		xs:decimal	Optional (attribute)
RollDragCoefficient		xs:decimal	Optional (attribute)
AirDragCoefficient		xs:decimal	Optional (attribute)
RadialDragCoefficient		xs:decimal	Optional (attribute)
ConstantPowerIntake		xs:decimal	Optional (attribute)
DeviceBatteryStoppingThreshold		xs:decimal	Optional (attribute)
EngineEfficiency		xs:decimal	Optional (attribute)
EnginePowerMax		xs:decimal	Optional (attribute)

Table 7. ICEV type

Name	Element Name	Type	Use
FuelType		xs:string	Required (attribute)

FuelTankCapacity		xs:decimal	Required (attribute)
FuelConsumptionRate		xs:decimal	Required (attribute)
EngineEfficiency		xs:decimal	Optional (attribute)
Torque		xs:decimal	Required (attribute)
ConstantPowerIntake		xs:decimal	Optional (attribute)
RollDragCoefficient		xs:decimal	Optional (attribute)
AirDragCoefficient		xs:decimal	Optional (attribute)
RadialDragCoefficient		xs:decimal	Optional (attribute)
EnginePowerMax		xs:decimal	Optional (attribute)

Table 8 : Info4Vehicle

Name	Element Name	Type	Use
Vehicle		VehicleType	Required [0,unbounded]
DefaultVehicle		DefaultVehicleType	Required [1,1]
FleetConfiguration		FleetConfigurationType	Optional [0,1]

You can review Info4Vehicle.xsd file for definition and details.

3.2.4 Info4Task

The Info4Task table containing the details of the tasks can be customized for different problems or purposes. It defines the routing task with demand and other information by extending the customer points defined in Info4Environment. The structure is given in Table 9.

Table 9 : Info4Task Table

Name	Element Name	Type	Use
Task			
	ProblemType	xs:string	Required [1,1]
	PerformanceMeasure	PerformanceMeasureType	Required [1,1]
	ShortestPath	ShortestPathType	Optional [0,1]
	VRP/TSP	VRP_TSPType	Optional [0,1]
	CEVRP	CEVRPType	Optional [0,1]
	DynamicRouting	DynamicRoutingType	Optional [0,1]
	...	...	...

PerformanceMeasure, which is a continuation of Info4Task and gives performance metrics, is given in Table 10, 11 and 12.

Table 10 : PerformanceMeasure type

Name	Element Name	Element Name	Type	Use
PerformanceMeasure			xs:complexType	Required [1,1]
	ObjectiveFunction		xs:complexType	Required [1,1]
		MinDistance	xs:string	Optional [0,1]

		MinTime	xs:string	Optional [0,1]
		MinEnergy	xs:string	Optional [0,1]
		MinTardiness	xs:string	Optional [0,1]
	MultiObjectiveFunciton		xs:complexType	Optional [0,1]
		Hybrid	xs:string	Optional [0,1]
	MultiCriteria		xs:complexType	Optional [0,1]
		Criteria	xs:string	Optional [0,1]

Table 11 : Task detail, ShortestPathType and VRP_TSPType

Name	Element Name	Element Name	Element Name	Type	Use
ShortestPath					Optional
	Origin			xs:complexType	Required [1,1]
		NodeId		xs:string	Required [1,1]
	Destination			xs:complexType	Required [1,1]
		NodeId		xs:string	Required [1,1]
VRP/TSP					Optional
	Customers				Required

		No		xs:string	Required [1,1]
		Name		xs:string	Optional [0,1]
		Location		LocationType	Required [1,1]
		Requests		xs:complexType	Required [1,1]
			Request	RequestType	Required [1,unbounded]
		CustomerPref		xs:string	Optional [0,1]
		PDType [P, D, PD]		xs:enumeration [xs:string]	Required [1,1]

Table 12 : Request Type table

Name	Element Name	Element Name	Type	Use
Request				
	ProductName		xs:string	Optional [0,1]
	ProductId		xs:string	Required [1,1]
	ReadyTime		xs:decimal/xs:dateTime	Optional [0,1]
	ServiceTime		xs:decimal/xs:dateTime	Optional [0,1]
	DueDate		xs:decimal/xs:dateTime	Optional [0,1]
	RequestTime		xs:decimal/xs:dateTime	Optional [0,1]

	LoadInformation		xs:complexType	Required [1,1]
		Quantity	xs:integer	Optional [0,1]
		Weight	xs:decimal	Optional [0,1]
		Volume	decimal	Optional [0,1]

You can review Info4Task.xsd file for definition and details.

3.2.5 Info4ChargingStation

Info4CS containing ChargeStation information and implementing the OCPP standard are given in Table 13.

Table 13 : Info4CS

Name	Element Name	Element Name	Type	Use
CountryCode			xs:string	Required
PartyId			xs:string	Required
Id			xs:string	Required
Publish			xs:boolean	Required
Name			xs:string	Required
Address			xs:string	Required
City			xs:string	Required
PostalCode			xs:string	Required
State			xs:string	Required
Country			xs:string	Required
ParkingType			xs:string	Optional
TimeZone			xs:string	Optional

ChargingWhenClosed			xs:boolean	Optional
LastUpdated			xs:datetime	Required
PublishAllowedTo			xs:complexTyp e	Optional
	Id		xs:string	Required
	Type		xs:string	Required
	VisualNumber		xs:string	Optional
	Issuer		xs:string	Optional
	GroupId		xs:string	Optional
Location			LocationType	Required
Evses			xs:complexTyp e	Required
	Evse		EvseType	Optional
Directions			xs:complexTyp e	Required
	Language		xs:string	Required
	Text		xs:string	Required
Operator			xs:complexTyp e	Optional
	Name		xs:string	Required
	Website		xs:string	Optional
	Logo	[url, thumbnail, category,type, width,height]	xs:complexTyp e	Optional
Owner			xs:complexTyp e	Optional
Facilities			xs:string	Optional
TimeZone			xs:string	Required
OpeningTimes			xs:complexTyp e	Optional

	Twentyfourseven			Optional
	RegularHours		xs:complexType	Optional
		Weekday	xs:integer	Optional
		PeriodBegin	xs:string	Optional
		PeriodEnd	xs:string	Optional
	ExceptionalOpenin		xs:string	Optional
Images			xs:string	Optional
EnergyMix			xs:complexType	Optional
	IsGreenEnergy		xs:boolean	Optional
	EnergySources		xs:complexType	Optional
		Source	xs:string	Optional
		Percentage	xs:integer	Optional
	SupplierName		xs:string	Optional
	EnergyProductName		xs:string	Optional

Table 14: EvseType

Name	Element Name	Element Name	Type	Use
Evse				
	Uid		xs:string	Required
	EvseId		xs:string	Required
	Status		xs:string	Required
	StatusSchedule		xs:complexType	Required
		PeriodBegin	xs:string	Required
		PeriodEnd	xs:string	Required

	Capabilities		xs:string	Required
	Connectors		xs:complexType	Required
		Id	xs:string	Required
		Standard	xs:string	Required
		Format	xs:string	Required
		PowerType	xs:string	Required
		MaxVoltage	xs:integer	Required
		MaxAmperage	xs:integer	Required
		MaxElectricPower	xs:integer	Required
		TariffIds	xs:string	Optional
		TermsAndConditions	xs:string	Optional
		LastUpdated	xs:string	Optional
	FloorLevel		xs:string	Optional
	Coordinates		xs:complexType	Optional
		Latitude	xs:decimal	Optional
		Longitude	xs:decimal	Optional
	PhysicalReference		xs:string	Optional
	Directions		xs:string	Optional
	ParkingRestrictions		xs:string	Optional
	Images	[Url, Thumbnail, Category, Type, Width, Height]	xs:complexType	Optional

3.2.6 Info4Algorithm

Info4Algorithm containing routing algorithm and parameters information. Standard are given in Table 1.

Table 15. Info4Algorithm

Name	Element Name	Element Name	Type	Use
Algorithm				

	name		xs:string	Required (attribute)
	Parameters		xs:sequence	Required [1, 1]
		Parameter		Required [1, unbounded]
		name	xs:string	Required [1,1]
		type	xs:enumeration (int, float, string, bool)	Required [1,1]

3.3 Outputs

Tables of the outputs from the algorithm are given in this section. Each output is given under the related section.

3.3.1 Route4Plan

Route4Plan is a data structure that contains the planned version of a route. This output may include solutions produced by optimization algorithms and provides detailed information about the determined route.

It can be used for the following purposes:

- Analysis and evaluation of planned routes
- Standardization of outputs of optimization algorithms
- Sharing of route plans between different systems

Route Plan structure is given in Table 12, 13, 14, 15, 16 and 17.

Table 16 : Solution type

Name	Element Name	Element Name	Type	Use
Solution				Required
	Name			Required

	ProblemType		xs:simpleType	Required
		VRP_TSP	xs:string	Optional
		CEVRP	xs:string	Optional
		DynamicRouting	xs:string	Optional
		. . .		
	ObjectiveFunction		xs:simpleType	Required
		MinDistance	xs:string	Optional
		MinTime	xs:string	Optional
		MinEnergy	xs:string	Optional
		MinTardiness	xs:string	Optional
	ChargingStrategy		xs:simpleType	Required
		FullCharge	xs:string	Optional
		PartialCharge	xs:string	Optional
		20-80	xs:string	Optional
	NumberOfCustomers		xs:integer	Required
	BatteryCapacity		xs:decimal	Required
	MaxSpeed		xs:decimal	Required
	VehicleMass		xs:decimal	Required
	MaximumLoadCapacity		xs:decimal	Optional
	MaximumLoadCapacityKg		xs:decimal	Optional

	BatteryRechargingRate		xs:decimal	Optional
	EnergyConsumptionRate		xs:decimal	Optional
	PerformanceMeasure		PerformanceMeasureTyp e	Require d
	Routes		xs:complexType	Require d
		Route	RouteType	Require d

Table 17 : RouteType

Name	Element Name	Element Name	Type	Use
Route				
	RouteId		xs:string	Required
	VehicleId		xs:string	Required
	DriverId		xs:string	
	PerformanceMeasure		PerformanceMeasureType	Required
	Nodes		xs:complexType	Required
		Node	NodeType	Required

Table 18 : Node type

Name	Element Name	Element Name	Type	Use
Node				
	NodeId		xs:string	Required
	NodeType		xs:simpleType	Required
		Depot	xs:string	Optional
		Customer	xs:string	Optional
		ChargeStation	xs:string	Optional

	RequestBatchId		xs:string	Optional
	Location		LocationType	Required
	ServiceInformation		ServiceInformationType	Required
	LoadInformation		LoadInformationType	Required
	PerformanceMeasure		PerformanceMeasureType	Required

Table 19: ServiceInformation type for Node

Name	Element Name	Type	Use
ServiceInformation			
	ReadyTime	xs:decimal xs:dateTime	Optional
	WaitingTime	xs:decimal xs:dateTime	Optional
	ServiceTime	xs:decimal xs:dateTime	Optional
	DueDate	xs:decimal xs:dateTime	Optional
	ArrivalTime	xs:decimal xs:dateTime	Optional

Table 20 : LoadInformation type for Node

Name	Element Name	Type	Use
LoadInformation			Required
	Weight	xs:decimal	Optional
	Volume	xs:decimal	Optional
	Units	xs:integer	Optional

Table 21 : PerformanceMeasure type for Node

Name	Element Name	Element Name	Type	Use
PerformanceMeasure				Required

	AccDistance		xs:decimal	Optional
	AccDuration		xs:decimal xs:dateTime	Optional
	AccEnergy		xs:decimal	Optional
	AccTardiness		xs:decimal xs:dateTime	Optional
	AccCost		xs:decimal	Optional
	RemainingCharge		xs:decimal	Optional
	RemainingCapacity		xs:complexType	Optional
		Volume	xs:decimal	Optional
		Weight	xs:decimal	Optional
	RemainingPayload			Optional
		Volume	xs:decimal	Optional
		Weight	xs:decimal	Optional

Acc: Accumilative

3.3.2 Route4Vehicle

RouteVehicle, which contains the route location information required for the vehicle, is given in Table 17.

Table 22 : Route4Vehicle

Name	Element Name	Element Name	Element Name	Type	Use
Route				xs:complexType	Required
	RouteId			xs:string	Required
	VehicleId			xs:string	Required
	RouteName			xs:string	Required
	StartPoint			xs:complexType	Required
		Id		xs:string	Required

		Location		xs:complexType	Required
			Latitude	xs:decimal	Required
			Longitude	xs:decimal	Required
			Elevation	xs:decimal	Required
		WayPoins		xs:complexType	Required
			Location[Latitude Longitude, Elevation]	xs:complexType	Required
	DeliveryPo ints			xs:complexType	Required
		Id		xs:string	Required
		Location		xs:complexType	Required
			Latitude	xs:decimal	Required
			Longitude	xs:decimal	Required
			Elevation	xs:decimal	Required
		WayPoints		xs:complexType	Required
			Location[Latitude , Longitude, Elevation]	xs:complexType	Required
	EndPoint			xs:complexType	Required
		Id		xs:string	Required
		Location		xs:complexType	Required
			Latitude	xs:decimal	Required
			Longitude	xs:decimal	Required
			Elevation	xs:decimal	Required
		WayPoints		xs:complexType	Required
			Location[Latitude , Longitude, Elevation]	xs:complexType	Required

3.3.3 Route4Sim

Route4Sim is used to test route plans in a simulation environment. It is suitable for integration with SUMO traffic simulation software. To create Route4Sim and its output, Map4Environment containing map information and cs.add.xml files containing stop information are needed. These files are kept as paths with sumoconfig.xml. The structure of the system to produce Route4Sim is given in Figure 2.

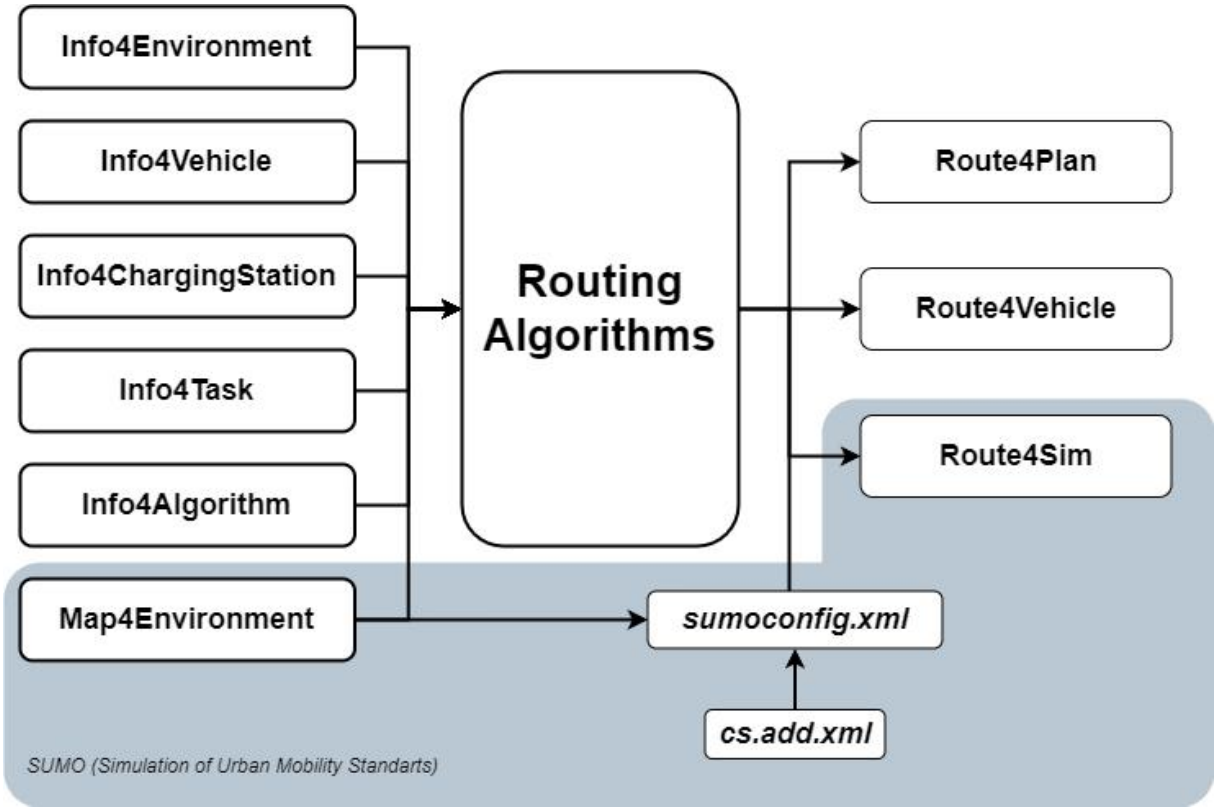

Figure 2: RoutingML diagram for Route4Sim

RouteSim complextypes are given in tables 19, 20 and 21 respectively. The structure of additional files is given in Appendix 5.3

Table 23 : Route4Sim

Name	Element Name	Type	Use
routes			

	vType	xs:complexType	Optional
	trip	xs:complexType	Required
	person	xs:complexType	Required
	vehicle	xs:complexType	Required

Table 24 : vType for RouteSim

Name	Element Name		Type	Use
vType			xs:complexType	Required
	param		xs:string	Optional
		key	xs:string	Required
		value	xs:string	Required
	id		xs:string	Required
	length		xs:decimal	Optional
	minGap		xs:decimal	Optional
	maxSpeed		xs:decimal	Optional
	vClass		xs:string	Optional
	emissionClass		xs:string	Optional
	accel		xs:decimal	Optional
	decel		xs:decimal	Optional
	sigma		xs:decimal	Optional

Table 25 : trip type for RouteSim

Name	Element Name	Type	Use
trip		xs:complexType	Optional
	id	xs:string	Required
	depart	xs:string	Required
	from	xs:string	Required

	to	xs:string	Required

Table 26 : person type for RouteSim

Name	Element Name	Element Name	Type	Use
person			xs:complexType	Optional
	id		xs:string	Required
	type		xs:string	Optional
	depart		xs:decimal	Required
	personTrip		xs:complexType	Optional
		from	xs:string	Required
		to	xs:string	Required
		lines	xs:string	Optional
	walk		xs:complexType	Optional
		edges	xs:string	Required

Table 27 : vehicle type for RouteSim

Name	Element Name		Type	Use
vehicle			xs:complexType	Required
	id		xs:string	Required
	type		xs:string	Required
	depart		xs:decimal	Required
	color		xs:string	Optional
	param		xs:complexType	Optional
		key	xs:string	Required
		value	xs:string	Required
	route		xs:complexType	Required
		edges	xs:string	Required

	stop		xs:complexType	Optional
		containerStop	xs:string	Optional
		chargingStation	xs:string	Optional
		until		Optional
		duration		Optional

4. APPENDIX

Routing Markup Language creates a standard and ensures that inputs and outputs are in specific formats. It makes the data universal. It can be used as an independent tool for new applications developed in this way. It makes it easier to test and make new developments. XML (eXtensible Markup Language) structure is an ideal basis for RML thanks to its hierarchical and flexible structure. In this section, information about XML will be given and RML structure will be explained.

4.1 XML

It is a format for storing data in a structured way, using tags. In RML, XML tags can be used to represent different components of the routing problem (e.g. vehicle, route, charging station). This makes it easier to read and standardize the data. An example xml structure is given inFigure 1.

```xml
<route>
  <vehicle>
    <id>1</id>
    <capacity>1000</capacity>
    <range>200</range>
  </vehicle>
  <nodes>
    <node id="1" x="10" y="20" />
    <node id="2" x="30" y="40" />
  </nodes>
  <edges>
    <edge source="1" target="2" distance="30" />
  </edges>
</route>
```

Figure 4 : XML example

- **<route>:** Represents a route.
- **<vehicle>:** Contains vehicle information (ID, capacity, range).

- **<nodes>:** Represents the nodes (points) on the route.
- **<edges>:** Represents the connections (edges) between nodes.

XSD schema is used to keep the tags used in XML in specific formats.

XSD : XML Schema Definiton is used to define the structure of XML documents. It ensures document validity and compatibility between different systems. An example xsd structure is given in Figure 2.

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="route">
    <xs:complexType>
      </xs:complexType>
  </xs:element>
  </xs:schema>
```

Figure 5 : XSD type example

4.2 XML Types

Type	Examples
xs:string	foo, bar, 123, 1a2b3c
xs:decimal	3.141, -123.78, 1.0E+10
xs:integer	42, -100, 0
xs:dateTime	2024-02-20T13:35:00, 2023-12-25T00:00:00+08:00

4.3 cs.add.xml and sumoconfig.xml

cs.add.xml and sumoconfig.xml files are xml structures with predefined schemas used in the SUMO simulation environment. You can review the links below to get detailed information about these structures.

https://sumo.dlr.de/xsd/additional_file.xsd

https://sumo.dlr.de/xsd/sumoConfiguration.xsd

https://sumo.dlr.de/xsd/routes_file.xsd

4.4 ROUTE XML Examples and Other Tools

https://sumo.dlr.de/docs/Tools/Net.html

Supporting Resources for Routing Markup Language (RoutingML) Integration

RoutingML Schema Documentation

Official schema definitions and XML structure references for all RoutingML components are available at:
https://pilestin.github.io/RoutingMarkupLanguage/html/schema.html

RoutingML Version Tracking

Version history and structural changes between releases (e.g., v1.03 → v1.04) can be explored via the version graph interface:
https://pilestin.github.io/RoutingMarkupLanguage/html/versions/v1.04.html

Task Visualization Tool

Interactive web tool for visualizing customer and depot locations used in dataset analysis and route planning validation:
https://pilestin.github.io/BuyukdereEVRouting/2-TaskVisualizer/index.html

Route Visualization Tool

Visualization platform for displaying Route4Vehicle (R4V) structures on map layers for comparative route analysis:
https://pilestin.github.io/BuyukdereEVRouting/4-RouteVisualizer/index.html

Fleet Route Analysis Dashboard

A web-based analytical interface for examining Route4Plan (R4P) outputs within a fleet management context:
https://pilestin.github.io/BuyukdereEVRouting/6-RouteAnalysis/index.html